

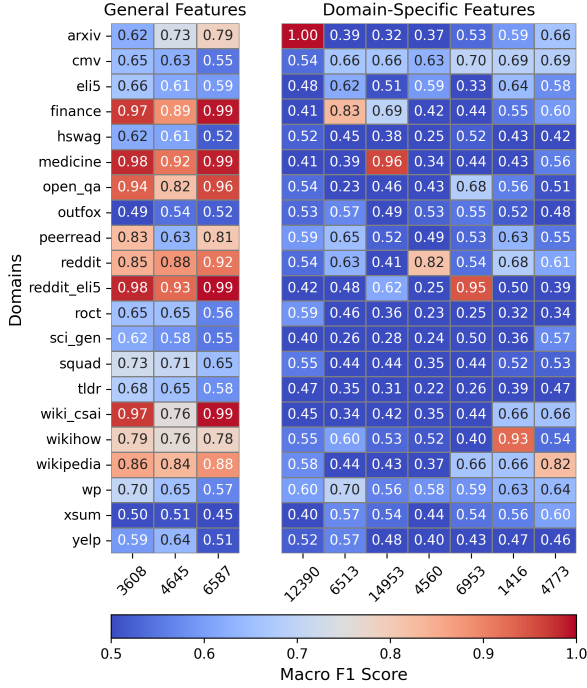


Figure 4: F1 Macro by the domains subsets for some general and domain-specific features for the 16 layer

on feature extraction can be found in Appendix D. Our analysis shows that features most susceptible to attacks and shallow text properties overlap minimally with those identified as important by XG-Boost. Specifically, features 8689 (detecting the GPT3.5+ family) and 14919 (detecting the Bloom family) are very sensitive to sentence length, while other distractions have limited impact on important features.

5 Important Features Interpretation

In this section, we discuss the insights from analyzing our feature interpretations (see details and examples in Appendix F), starting with the most robust features: 3608, 4645, 6587, 8264, and 14161. Their performance in the ATD task across various domains and models is shown in Figures 4 and 16.

Strong activations of these features correlate with common LLM-generated text characteristics, such as excessive complexity (3608), assertive claims (4645), wordy introductions (6587), repetition (8264), and formality (14161). These features perform well on GPT3.5+ and other modern LLMs like LLaMa and Gemma, especially for domains like finance, medicine, and Wiki-CSAi. However, texts from arXiv are less distinguishable, suggesting GPT models mimic scientific writing more closely.

Feature 8264 stands out with near-perfect perfor-

mance for GPT3.5+, controlling the conciseness vs. repetition of concepts. Older models lack this feature, leading to lower detectability.

Domain-specific features include overcomplicated syntax (arXiv, feature 12390), excessive details (finance, feature 6513), speculative links (Reddit, feature 4560), and hallucinated facts (Wikipedia, feature 4773). Improper tone (medicine, feature 14953) also signals machine-generated texts.

The most challenging domains for detection are Outfox (essays) and Yelp (reviews), where models mimic human-like writing. This suggests that general “overcomplexity” features may not be effective when models are instructed to avoid such traits.

6 Conclusion

Our analysis shows that modern LLMs often generate easily detectable text due to specific writing styles, such as long-winded introductions, excessive synonym substitution, and repetition. However, adversaries can bypass these features by using less formal, more personalized prompts, like student essays, leading to more human-like outputs.

Unlike previous approaches, we perform a multifaceted analysis of features for Artificial Text Detection (ATD). We select key features, examine their behavior across domains and generators, and interpret them both through extreme values (manual) and medium shifts (steering + LLM interpretation). This approach provides deeper insights into feature meanings. For example, our interpretation of feature 3608 contrasts with Neuropedia’s narrow view, which links it to “tokens associated with mathematical expressions.” Similarly, feature 4645, described by Neuropedia as related to “key-words on diabetes,” is more broadly relevant in our analysis.

We conclude that Sparse Autoencoder-based analysis of ATD datasets is a valuable tool for understanding text generators, detectors, and how detectors generalize to new setups. Our findings highlight that detecting AI-generated text is easy with a default prompt but becomes difficult when prompt style changes, a crucial consideration for ATD developers.

7 Limitations

Artificial text detection (ATD) is a highly complex and evolving task. With new LLMs emerging almost every month, it is difficult to predict how our

method will perform on future artificial text generators. Additionally, novel attack strategies continue to appear, and our approach covers only a subset of them. Besides, some of SAE features we studied remain challenging to interpret, as not all exhibit clear semantic meaning.

Finally, in this short paper, we used a single Sparse Autoencoder (SAE) on the residual stream of Gemma 2-2B. Exploring different SAEs on other LLMs could reveal new features and offer additional insights into artificial text detection. We leave this for future work.

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